Georeferencing: Castra Vetera

geo0240 · a photographed map and a reconstruction drawing in today's coordinates



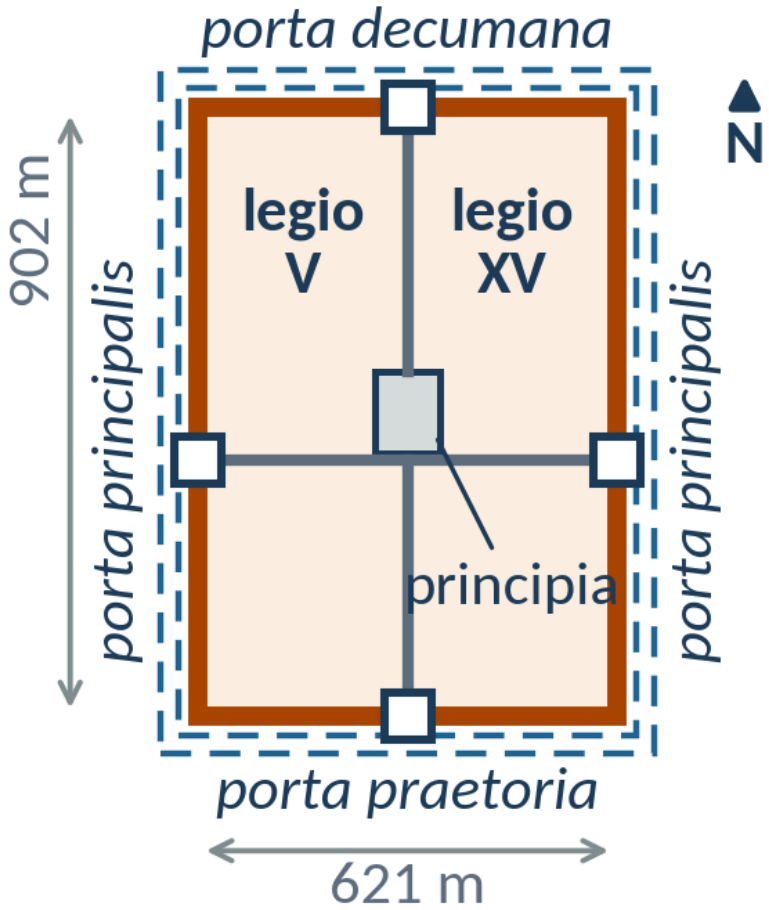
Historical background

Vetera I · AD 69/70 · Colonia Ulpia Traiana



Castra Vetera I on the Fürstenberg

- ***castra*** (Latin, plural) = military camp; *castra legionis* = legionary fortress; *castellum* = auxiliary fort; not a "castle"
- **13/12 BC:** Augustan base opposite the Lippe mouth, probably for Drusus' campaigns; earth-and-timber camps
- **AD 14–60:** two legions, rebuilt several times
- **c. AD 60: double-legion fortress** (legio V and XV), 902 m × 621 m ≈ 56 ha, the largest known
- **Lower Germanic Limes:** river frontier (*ripa*), no continuous wall; UNESCO 2021



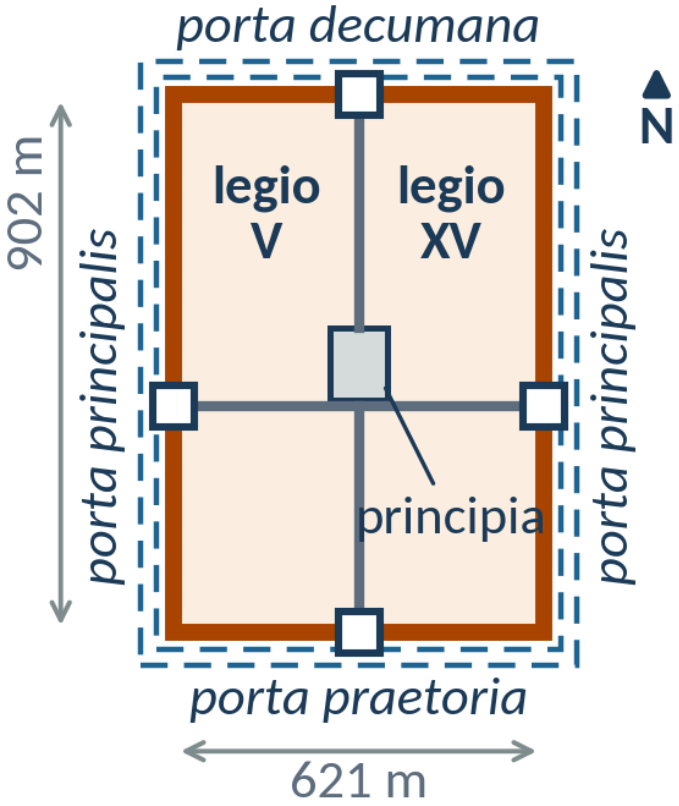
schematic: wall circuit to scale, ditches exaggerated

Wikipedia: Vetera · Lower Germanic Limes · Castra · Vetera (de)

Text: distilled from Wikipedia (CC BY-SA 4.0), see the last slide. Figures: own work (make_slides.py), CC BY-SA 4.0.

Vetera I, c. AD 60: defences and layout

Latin	English	Vetera I, c. AD 60
vallum	rampart, palisade	earlier phases: earth-and-timber
murus	wall, wall walk	3 m, timber frame, clay infill
turres	towers	presumed, not proven
portae	gates	4, each 2 towers + bridge
fossae	V-shaped ditches	2, plus abatis
principia	headquarters	stone, 120 m × 94.8 m
valetudinarium	hospital	stone, 83.5 m × 83.5 m



Wikipedia: Vetera · Castra · Vetera (de)

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AD 68–70: four emperors, Batavian revolt

Rome: four emperors in one year

	emperor	end
68	Nero †	9 June 68
68–69	Galba	murdered 15 Jan 69
69	Otho	suicide, April 69
69	Vitellius	acclaimed by the Rhine legions; killed 20 Dec 69
69–79	Vespasian	Flavian dynasty

Lower Rhine: revolt of the Batavi

- **Batavi:** Germanic people in the Rhine delta, allies supplying auxiliary troops
- **Julius Civilis:** Batavian noble, Roman citizen, prefect of a cohort
- **69:** revolt during the civil war; Vetera besieged
- **70:** Vetera surrenders (hunger), legions V and XV destroyed, fortress burnt
- **summer 70:** Cerialis defeats Civilis at Vetera

Wikipedia: Year of the Four Emperors · Revolt of the Batavi · Gaius Julius Civilis

Text: distilled from Wikipedia (CC BY-SA 4.0), see the last slide. Figures: own work (make_slides.py), CC BY-SA 4.0. Main ancient source: Tacitus, Histories.

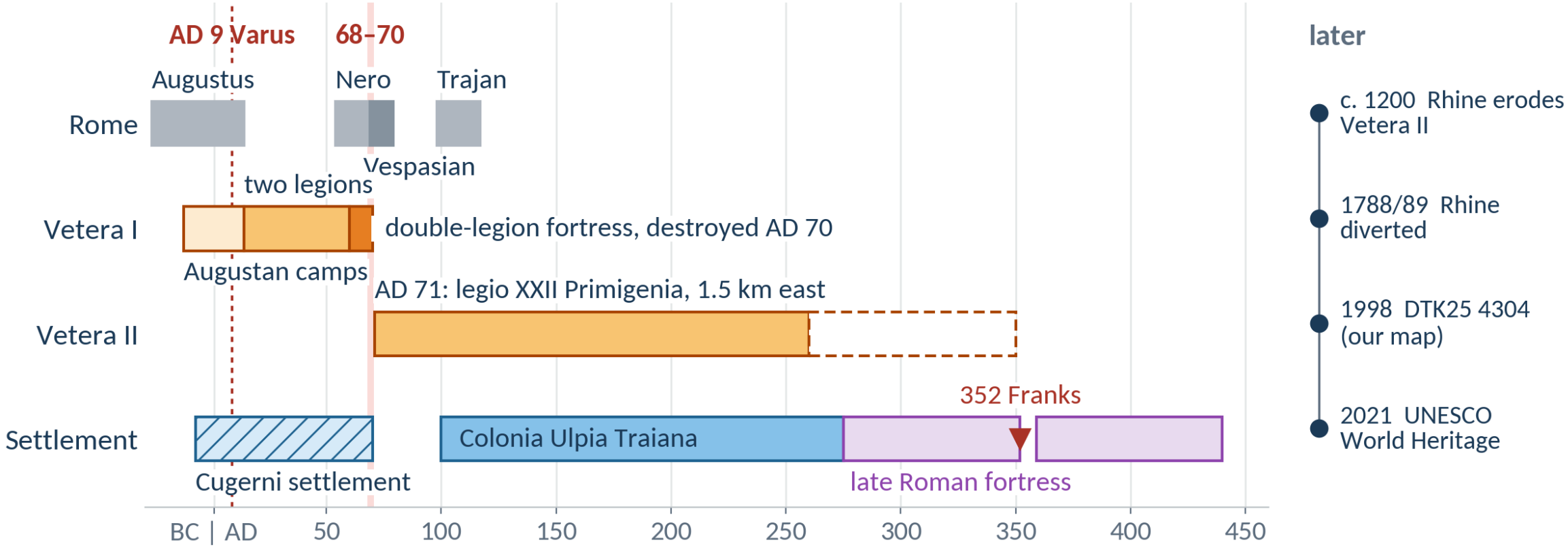
Colonia Ulpia Traiana (Xanten)

- **Before:** settlement of the Cugerni next to Vetera I, probably burnt in AD 70
- **c. AD 100:** Emperor Trajan (Marcus *Ulpius Traianus*) grants the rank of *colonia*, the highest rank of a Roman town
- **73 ha** inside a 3.4 km town wall with 22 towers; c. 10 000 inhabitants; grid of 40 *insulae* with forum, capitol, baths, harbour temple, amphitheatre; harbour on a Rhine arm
- Built partly with stones from the ruins of Vetera I
- **3rd–5th century:** decline; late Roman fortress on the 9 central insulae; taken by the Franks in 352, abandoned in the 5th century
- **Today:** LVR Archaeological Park Xanten; part of the UNESCO World Heritage site since 2021. Find it on the DTK25 photo!

Wikipedia: Colonia Ulpia Traiana · Xanten · Archaeological Park Xanten

Text: distilled from Wikipedia (CC BY-SA 4.0), see the last slide. Figures: own work (make_slides.py), CC BY-SA 4.0.

Xanten in Roman times: a time scale



Wikipedia: Vetera · Colonia Ulpia Traiana · Lower Germanic Limes

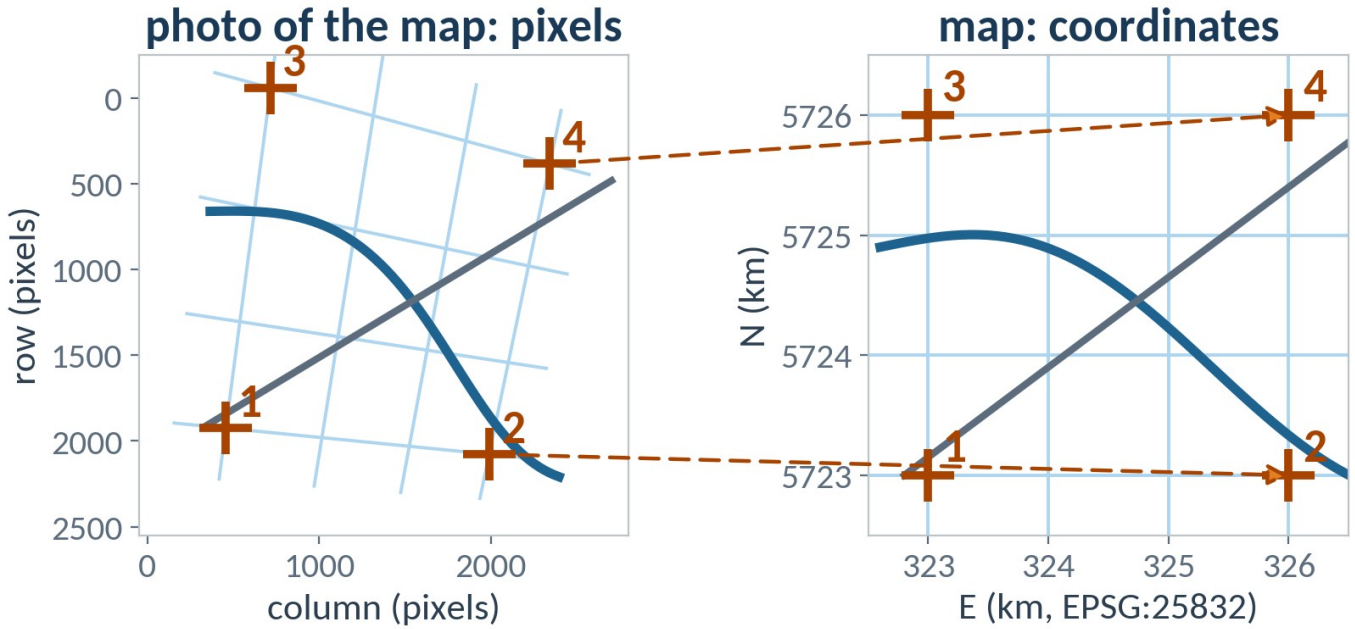
Text: distilled from Wikipedia (CC BY-SA 4.0), see the last slide. Figures: own work (make_slides.py), CC BY-SA 4.0.

The method

control points · transformations · RMS · resampling



What georeferencing does



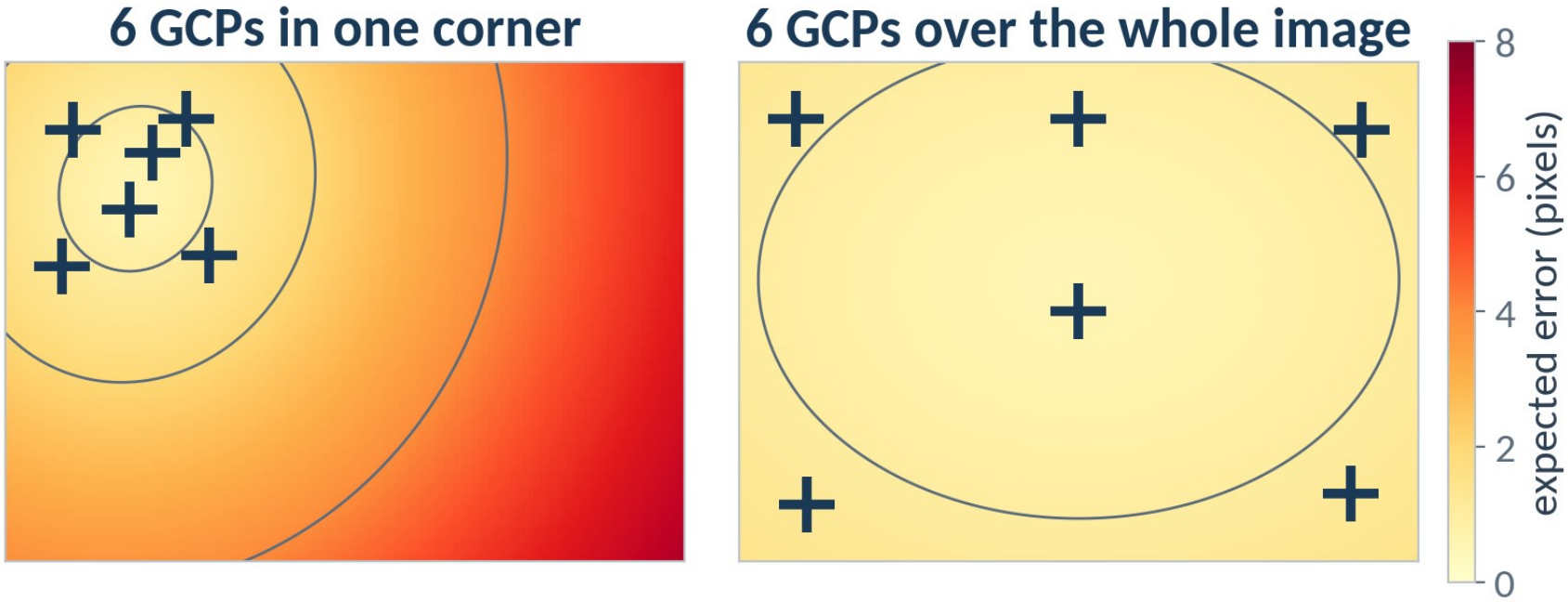
- **Ground control point (GCP):** known in the image (column, row) and on the map (E, N)
- **Transformation:** fitted to the GCPs, applied to every pixel → GeoTIFF in the target CRS

[QGIS user manual: Georeferencer]

Wikipedia: Georeferencing · Ground control point · Image registration · World file · GeoTIFF

Figure: own work (make_slides.py), CC BY-SA 4.0; schematic.

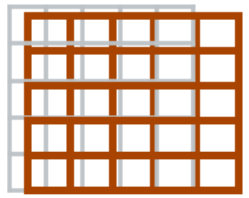
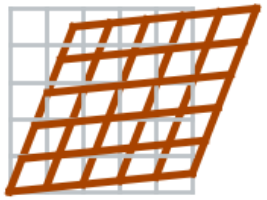
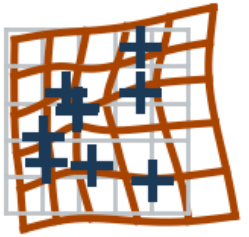
Ground control points: where to click



- **Spread** the GCPs over the whole image, corners included: no extrapolation
- **Sharp and stable:** grid crossings, road crossings, bridges; not rivers or forest edges
- **Enough:** 2–3 × the minimum of the transformation, plus check points

Figures: own work (make_slides.py), CC BY-SA 4.0; simulated example, not the map photo.

Transformations in the Georeferencer

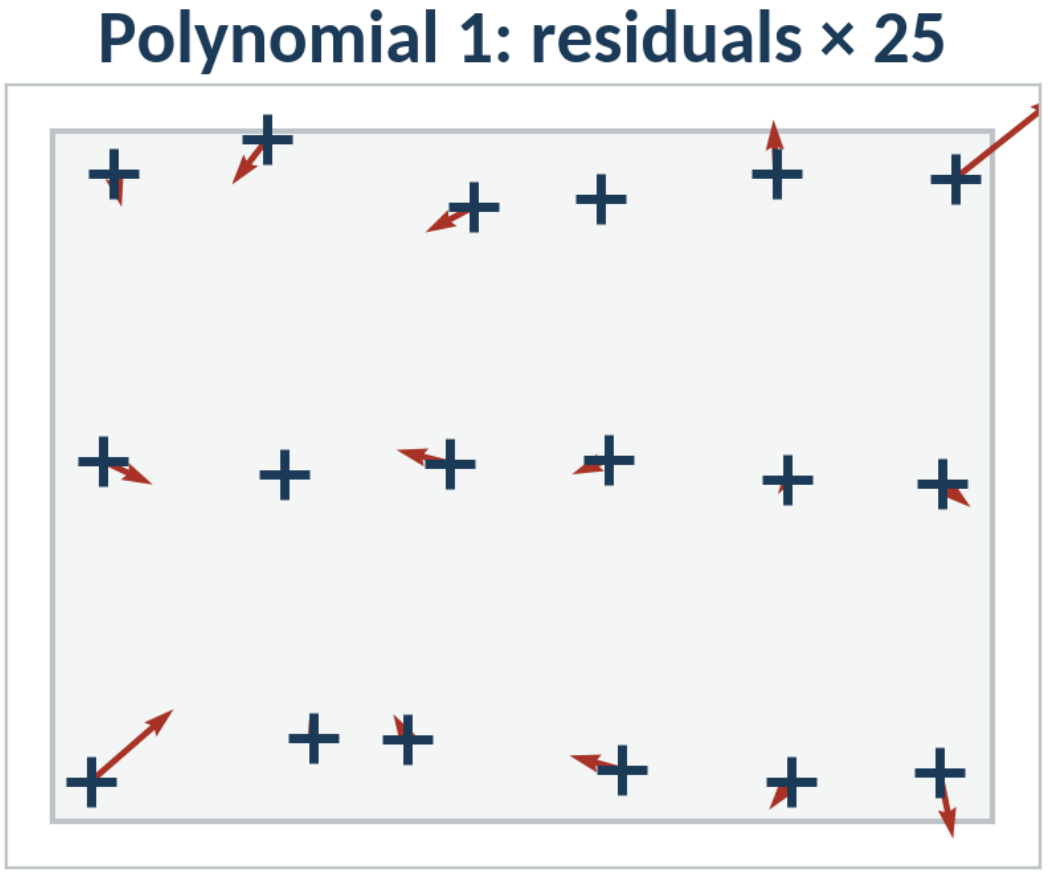
Linear	Helmert	Polynomial 1	Projective	Polynomial 2	Thin Plate Spline
					
shift + scale ≥ 2 GCPs	+ rotation ≥ 2 GCPs	+ shear (affine) ≥ 3 GCPs	tilted photo ≥ 4 GCPs	bends lines ≥ 6 GCPs	rubber sheet ≥ 10 GCPs

- **More parameters** fit the GCPs better, but need more GCPs and can bend the image between them
- **Choose by the source:** scan of a good map → Helmert / Polynomial 1; photo taken at an angle → Projective; warped paper → Polynomial 2; poor originals → Thin Plate Spline

Wikipedia: Helmert transformation · Affine transformation · Homography · Thin plate spline

Text: QGIS user manual, Georeferencer (docs.qgis.org, CC BY-SA 3.0). Figures: own work (make_slides.py), CC BY-SA 4.0.

Residuals and RMS error



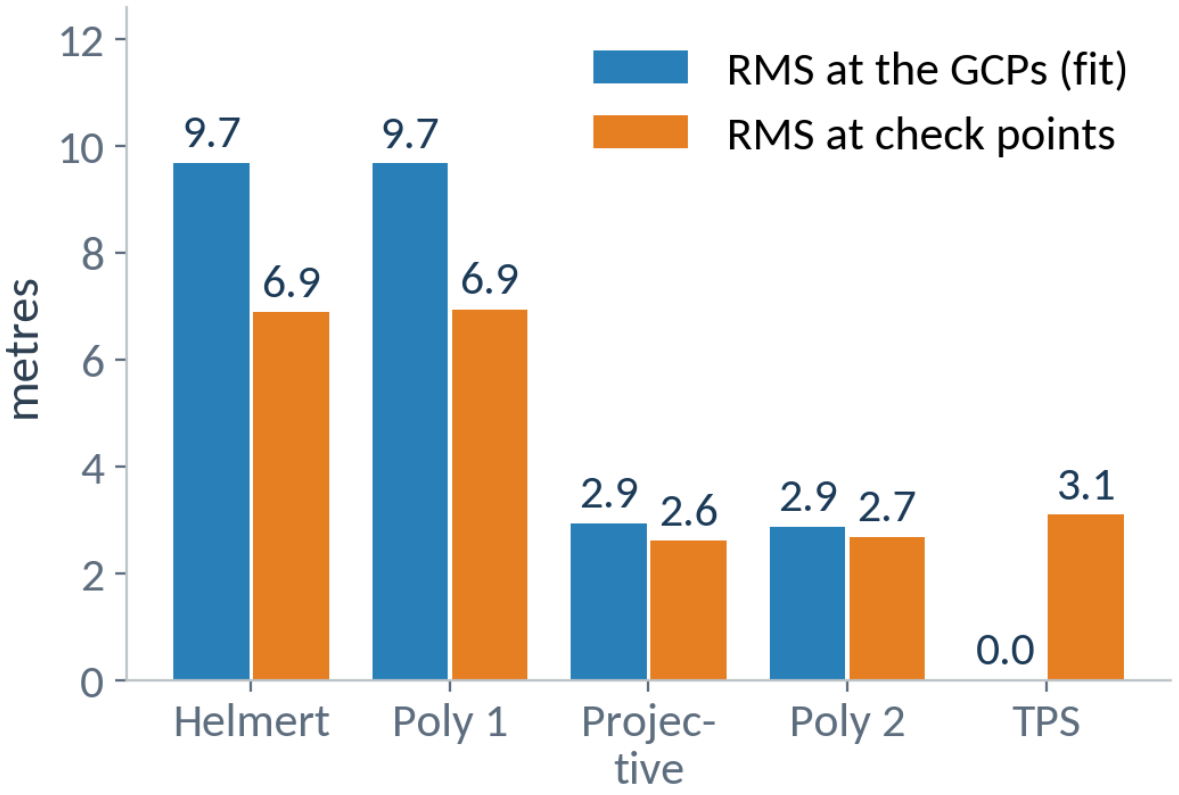
$$RMS = \sqrt{[\ \Sigma\ (dX^2 + dY^2)\ /\ n\ ]}$$

- **Residual:** how far a GCP lands from its map position
- **RMS:** one number for all GCPs, in pixels or metres
- **Systematic pattern** → transformation too simple
- **One long arrow** → wrong click: fix it or switch the GCP off

Wikipedia: Errors and residuals · Root-mean-square deviation · Least squares

Figures: own work (make_slides.py), CC BY-SA 4.0; simulated example, not the map photo.

A small RMS is not yet an accurate map

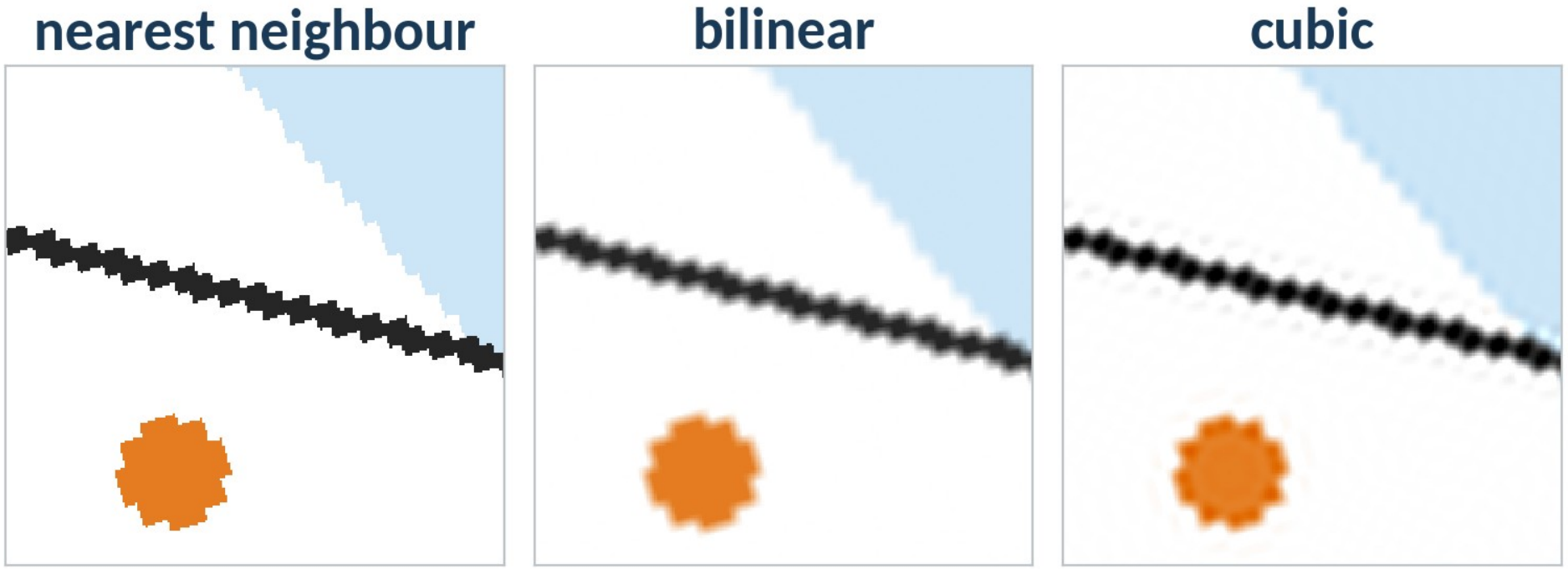


- **RMS at the GCPs** shrinks with every extra parameter
- **Thin Plate Spline (TPS):** RMS = 0 by construction
- **Check points:** switch 3–5 GCPs off and look at their error
- **Report:** transformation, number of GCPs, RMS and check error in metres

Wikipedia: Overfitting · Cross-validation

Figures: own work (make_slides.py), CC BY-SA 4.0; simulated example, not the map photo.

Resampling: new pixels from old ones



- **Nearest neighbour:** keeps the original values; required for classes and codes
- **Bilinear, cubic:** smoother, but mixed values; fine for photos and map scans

Wikipedia: Nearest-neighbor interpolation · Bilinear interpolation · Bicubic interpolation

Figure: own work (make_slides.py), CC BY-SA 4.0; a synthetic map patch rotated by 17°, ×5.

Georeferencing step by step

```
import numpy as np, pandas as pd
g = pd.read_csv("dtk_photo.points", comment="#")   # QGIS GCP file
A = np.c_[g.sourceX, g.sourceY, np.ones(len(g))]    # Polynomial 1
xy = g[["mapX", "mapY"]].to_numpy()
coef, *_ = np.linalg.lstsq(A, xy, rcond=None)
print("RMS", np.sqrt(((A @ coef - xy) ** 2).sum(1).mean()))
```

```
gdal_translate -gcp 103.1 402.4 322000 5728000 ... a.jpg a_gcp.tif
gdalwarp -t_srs EPSG:25832 -order 1 -r cubic a_gcp.tif a_utm.tif
# -order 2: Polynomial 2,  -tps: Thin Plate Spline
```

Same mathematics as the Georeferencer: least squares on the GCPs, then warp and resample.
Extend: leave-one-out check error.

Code: own example (CC BY-SA 4.0); GDAL utilities gdal_translate, gdalwarp.



Take-aways: georeferencing

- **Know the CRS of your GCPs:** read the map legend (grid, datum)
- **Spread sharp, stable GCPs** over the whole image
- **Choose the transformation by the source,** not by the smallest RMS
- **Judge with check points** and by overlaying a reference map
- **Resample with nearest neighbour** when the values are classes; cubic for photos
- **Keep the .points file:** it documents and repeats your work

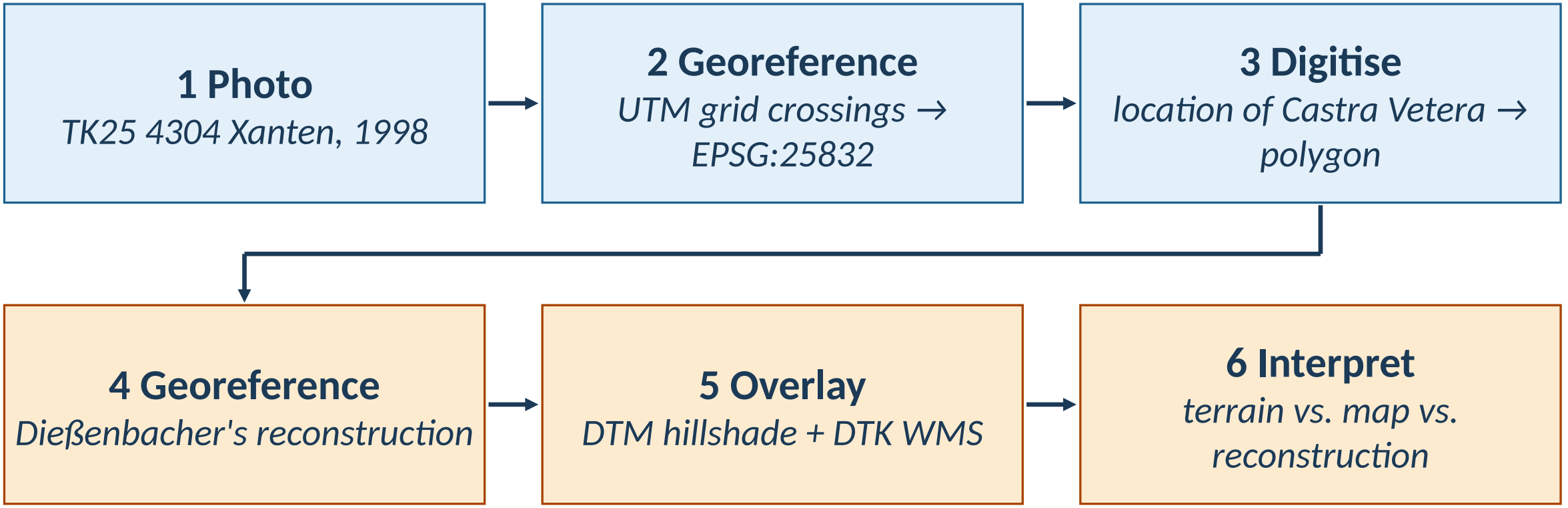


Case study: Castra Vetera

TK25 4304 Xanten (1998) · Vetera I reconstruction · DTM



The processing chain



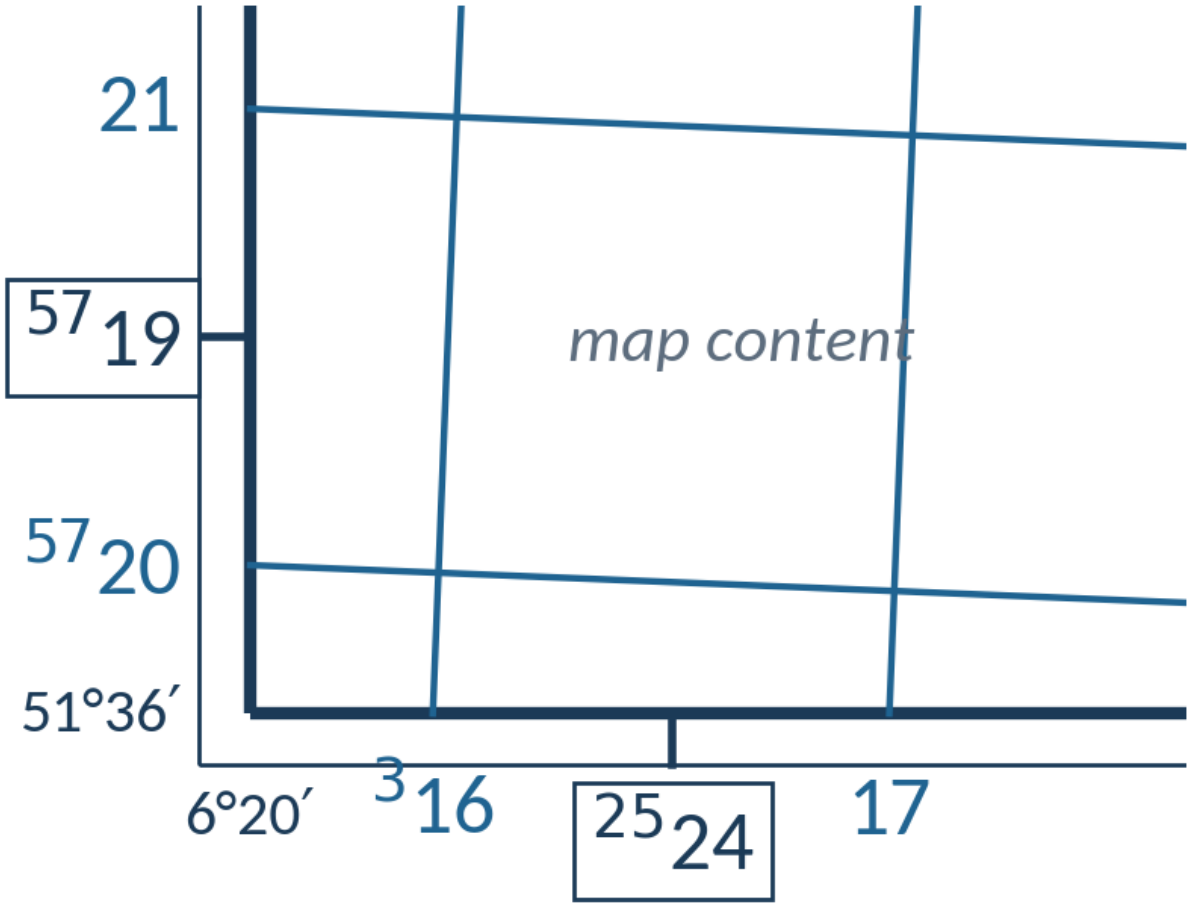
Tool: QGIS Georeferencer (Layer → Georeferencer…), digitising in a GeoPackage

Investigate the map

- **Type of map:** which map series, scale, edition name?
- **Publisher:** who made it? Who is responsible today?
- **Year:** edition, last update, sales period – how current is the content?
- **CRS:** geodetic datum, ellipsoid, projection of the map content?
- **Grids:** which coordinate grids are printed, and in which CRS? Which EPSG code do you use in the Georeferencer?
- **Look at:** cover, legend (Geodätische Grundlagen, Koordinaten und Höhen der Karte), map margin



Read the map margin



- Which lines belong to the UTM grid?
- What do the numbers mean? What are the small raised digits?
- What is the interval between two lines – in km and in cm on the map?
- And the boxed numbers and the corner values?

Wikipedia: Universal Transverse Mercator · Gauss-Krüger (de)

Figure: own schematic after the map margin (not to scale), CC BY-SA 4.0.

Our map has three coordinate systems

on the map sheet	datum	EPSG	for the GCPs
UTM zone 32, km grid	ETRS89 (“WGS84/ETRS89”)	25832	yes: grid crossings
Gauss-Krüger, zone 2	DHDN (Potsdam, Bessel)	31466	only with 31466
geographic, ° ′ ″	DHDN (Potsdam, Bessel)	4314	never as 4326

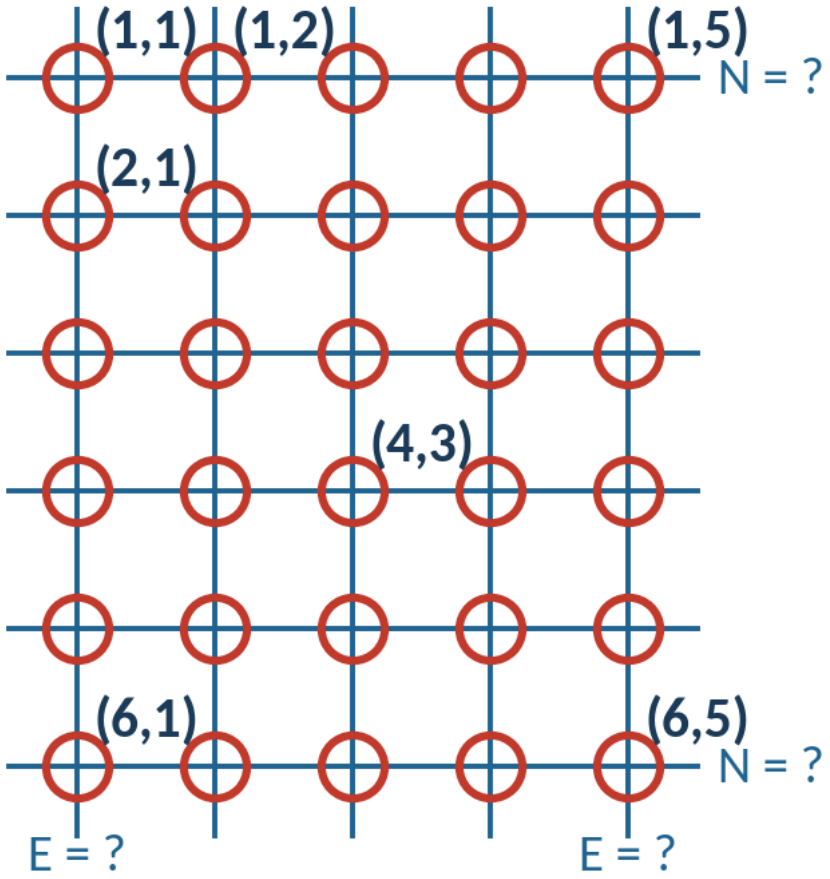
- **Read the legend first:** which grid, which datum? The sheet itself says how to convert GNSS positions: +2.5″ longitude, +4.8″ latitude
- **Same numbers, wrong datum:** about 155 m off (geo0220, DHDN → ETRS89)
- **Our choice:** UTM grid crossings at full kilometres, target CRS EPSG:25832

Wikipedia: Universal Transverse Mercator · Gauß-Krüger-Koordinatensystem (de) · Deutsches Hauptdreiecksnetz (de)

Source: legend of DTK25 4304 Xanten, Landesvermessungsamt NRW, 1998 (facts only).



Ground control points: the grid crossings



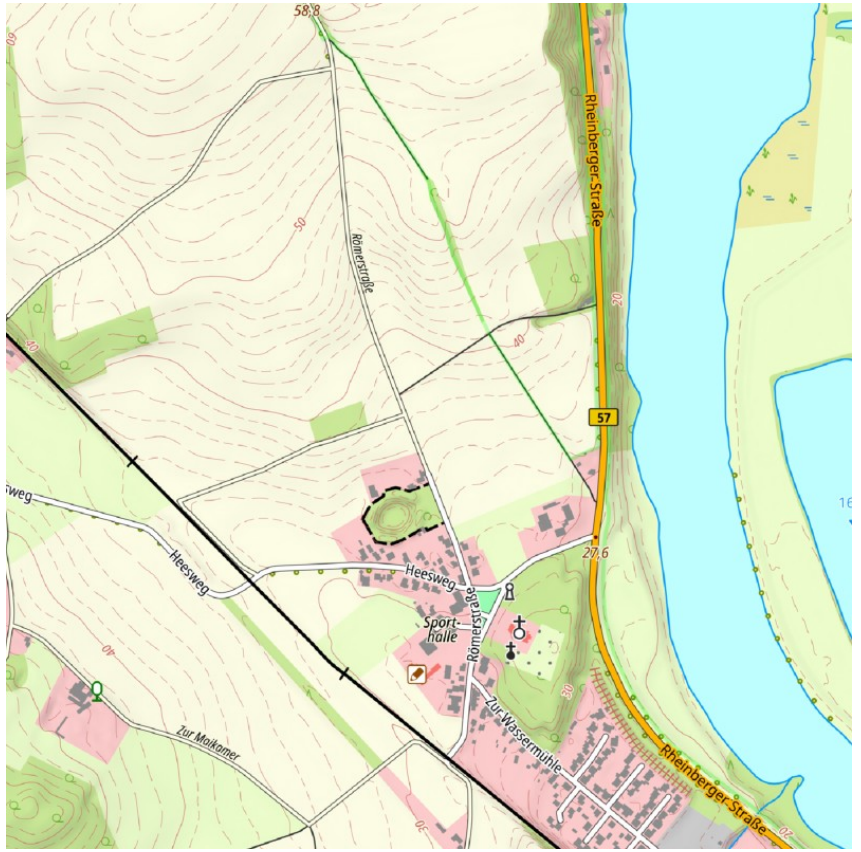
- The red circles on the detail number the crossings (row, column): a common reference for everyone
- Determine (E, N) of every crossing from the blue numbers in the margin
- Enter them in the Georeferencer, target CRS EPSG:25832
- Compare transformations and their RMS error

Figure: own work, CC BY-SA 4.0.

Today's DTK: where is Castra Vetera?



1:25 000



about 1:6 000 (zoomed in)

- 1998: label and KD mark the approximate location
- What is left today?
- Digitise the location from the 1998 map

© Geobasis NRW (dl-de/zero-2-0), WMS. Left 2.8 km × 2.8 km, right 1.4 km × 1.4 km, EPSG:25832.

Task: read the DTM hillshade



- Overlay your georeferenced map and the polygon of Castra Vetera with the DTM hillshade
- Which pronounced features does the hillshade show around the polygon? Describe them
- Which of them do you find on the 1998 map, which on today's DTK?

Wikipedia: Hillshading · Digital elevation model

DTM hillshade: © Geobasis NRW (dl-de/zero-2-0), WMS (wms_nw_dgm-schummerung), 2.8 km × 2.8 km.

Task: interpret the reconstruction

- Georeference Dießenbacher's reconstruction (GCPs from your georeferenced map and the DTM)
- Overlay it with the DTM hillshade and the DTK
- How does the reconstruction imagine the surroundings of Castra Vetera?
- Find inconsistencies between the reconstruction and the terrain. What could be the reasons?



Sources and further reading

Castra Vetera I on the Fürstenberg: Vetera · Lower Germanic Limes · Castra · Vetera (de)

Vetera I, c. AD 60: defences and layout: Vetera · Castra · Vetera (de)

AD 68–70: four emperors, Batavian revolt: Year of the Four Emperors · Revolt of the Batavi · Gaius Julius Civilis

Colonia Ulpia Traiana (Xanten): Colonia Ulpia Traiana · Xanten · Archaeological Park Xanten

Xanten in Roman times: a time scale: Vetera · Colonia Ulpia Traiana · Lower Germanic Limes

What georeferencing does: Georeferencing · Ground control point · Image registration · World file · GeoTIFF

Transformations in the Georeferencer: Helmert transformation · Affine transformation · Homography · Thin plate spline

Residuals and RMS error: Errors and residuals · Root-mean-square deviation · Least squares

A small RMS is not yet an accurate map: Overfitting · Cross-validation

Resampling: new pixels from old ones: Nearest-neighbor interpolation · Bilinear interpolation · Bicubic interpolation

Read the map margin: Universal Transverse Mercator · Gauss-Krüger (de)

Our map has three coordinate systems: Universal Transverse Mercator · Gauß-Krüger-Koordinatensystem (de) · Deutsches Hauptdreiecksnetz (de)

Task: read the DTM hillshade: Hillshading · Digital elevation model

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